# ECE343 Spring 2025

## Objective 5.1: Coloumb's Law



| Lesson | Topics | Reading | Notes |
|---|---|---|---|
| 1 | Course Overview & Introduction | Ch.1 | LO 1.1-1.2 |
| **Chapter 2: Electric and Magnetic Fields** | | | |
| 2 | Fields, Parameters, Field Quantities | Ch. 2 | LO 2.1-2.2 |
| **Chapter 3: Transmission Lines** | | | |
| 3 | Lumped Element Model | 3.1-3.4 | LO 3.1 |
| 4 | Telegrapher's Equations | 3.5-3.6 | LO 3.1 |
| 5 | Characteristic Impedance & Wave Propagation | 3.7-3.9 | LO 3.2 |
| 6 | Coaxial, Reflection Coefficient, VSWR | 3.10,3.12-3.14 | LO 3.2-3.3 |
| 7 | Terminated Transmission Lines | 3.15,3.17-3.19 | LO 3.4 |
| 8 | Impedance Matching | 3.21-3.23 | LO 3.4 |
| 9 | Impedance Matching | 3.21-3.23 | LO 3.4 |
| 10 | Smith Charts | Supplemental | LO 3.5 |
| 11 | Smith Charts | Supplemental | LO 3.5 |
| 12 | Smith Charts | Supplemental | LO 3.5 |
| 13 | Smith Charts | Supplemental | LO 3.5 |
| 14 | Transients | Supplemental | LO 3.6 |
| 15 | Flex Day | | |
| 16 | Reflection/Transmission Coefficient Lab | Lab Packet | LO 3.7 |
| 17 | Reflection/Transmission Coefficient Lab | Lab Packet | LO 3.7 |
| **Chapter 4: Vector Analysis** | | | |
| 18 | Vector & Coordinate System Review | Ch. 4 | LO 4.1-4.4 **GR 1 due** |
| **Chapter 5: Electrostatics** | | | |
| 19 | Coulomb's Law | 5.1-5.3 | LO 5.1 |
| 20 | Surface and Volume Charge | 5.4 | LO 5.1 |
| 21 | Gauss' Law (Integral Form) | 5.5-5.6 | LO 5.2 |
| 22 | Gauss' Law (Differential Form) | 5.7-5.8 | LO 5.2 |
| 23 | Electric Potential | 5.9-5.10, 5.15 | LO 5.3 |

| Lesson | Topics | Reading | Notes |
|---|---|---|---|
| 24 | BVP (Capacitor, Coax), Electric Potential | 5.16 | LO 5.3 |
| 25 | Electric Boundary Conditions | 5.18-5.22 | LO 5.4 |
| 26 | BC's/Capacitance | 5.23-5.24 | |
| 27 | Electrostatic Lab | Lab packet | |
| **Chapter 6: Steady Current & Conductivity** | | | |
| 28 | Current density, Conductivity, Resistance | Ch. 6 | |
| **Chapter 7: Magnetostatics** | | | |
| 29 | Magnetic Fields, Gauss' Law for Magnetism | 7.1-7.3 | LO 7.1-7.2 **GR 2** |
| 30 | Ampere's Law | 7.4-7.5, 7.9 | LO 7.3 |
| 31 | Ampere's Law Applied | 7.6-7.8 | LO 7.3 |
| 32 | Magnetic Boundary Conditions | 7.10-7.14 | LO 7.4 |
| 33 | Magnetostatics Lab | Lab packet | |
| **Chapter 8: Time-Varying Fields** | | | |
| 34 | Faraday's Law, Displacement Current, Maxwell Eqns | 8.3, 8.8, 8.9 | LO 8.1 |
| 35 | Faraday's Law, Displacement Current, Maxwell Eqns | 8.3, 8.8, 8.9 | LO 8.2-8.3 |
| **Chapter 9: Plane Waves in Lossless Media** | | | |
| 36 | Maxwell's Equations in lossless, source-free region | 9.1-9.2 | LO 9.1 |
| 37 | Maxwell's Equations in lossless, source-free region | 9.1-9.2 | LO 9.1 |
| 38 | Uniform Plane Waves & Polarization | 9.4-9.6 | LO 9.2-9.3 |
| 39 | Normal Incidence | Vol 2, 5.6-5.7 | LO 9.4-9.5 |
| 40 | Oblique Incidence | Vol 2, 5.6-5.7 | LO 9.5 |
| 41 | Oblique Incidence | Vol 2, 5.6-5.7 | LO 9.5 |
| 42 | Incidence Lab | TBD | LO 9.5 |

# Ch 5 Learning Objectives

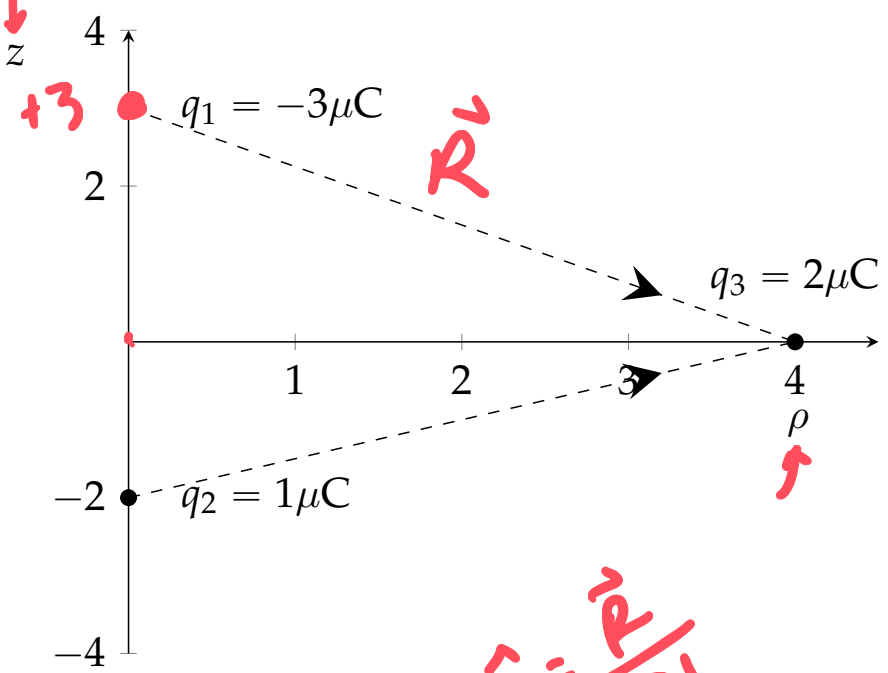
5.1 I can apply Coloumb's Law to use direct integration to calculate the electric field due to charge density in varoius geometries, including rectangular, cylindrical, and spherical coordinate systems.

5.2 I can apply the integral and differential forms of Gauss' Law to calculate the electric field due to charge density in various geometries, including rectangular, cylindrical, and spherical coordinate systems.

5.3 I can calculate the electric field from the scalar electric potential in a variety of geometries.

5.4 I can apply the electric field boundary conditions to calculate electric fields on both sides of a material boundary.

# Coulomb's Coloumb Force




$$\vec{F}_{mn} = \frac{q_m q_n}{\epsilon 4\pi |R|^2}\hat{R} = \frac{q_m q_n}{\epsilon 4\pi |R|^3}\vec{R}$$

with: $\vec{R} = \vec{r} - \vec{r}\,'$

① Draw a picture ✓

② Write Coulomb's Law

$$\vec{F}_T = \vec{F}_{13} + \vec{F}_{23}$$

$$\vec{F}_{13} = \frac{8_1 8_3}{4\pi\epsilon |R|_{13}^3}\vec{R}_{13}$$

$$\vec{F}_{23} = \frac{8_2 8_3}{4\pi\epsilon |R_{23}|^3}\vec{R}_{23}$$

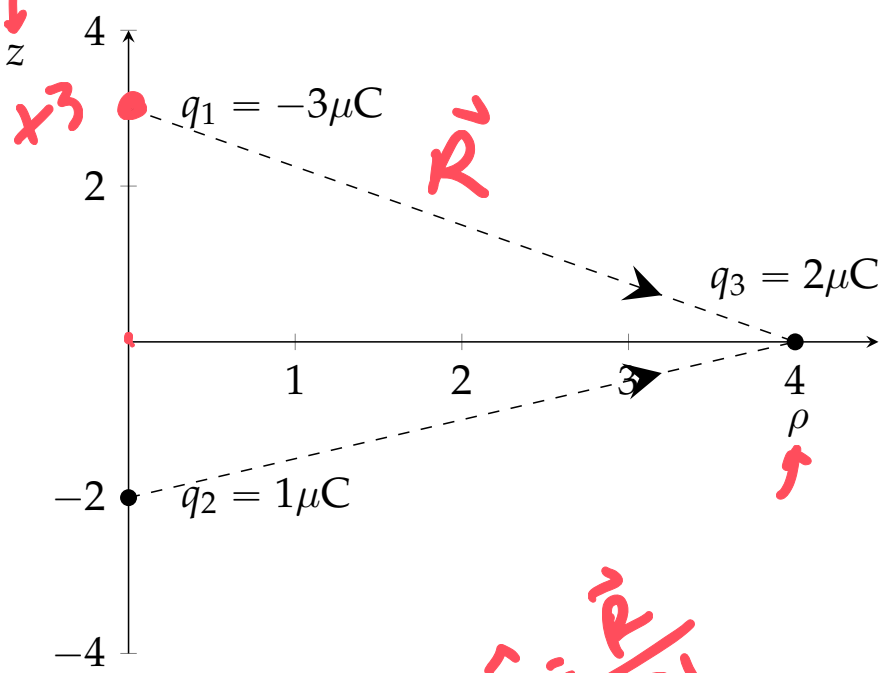
③ Define quantities

$$R_{13} = \hat{\rho}(4-0) + \hat{z}(0-3) = \hat{\rho}4 - \hat{z}3$$

$$|R| = \sqrt{4^2 + 3^2} = 5$$

④ Calc.!

# Coloumb Force




$$\vec{F}_{mn} = \frac{q_m q_n}{\epsilon 4\pi |R|^2}\hat{R} = \frac{q_m q_n}{\epsilon 4\pi |R|^3}\vec{R}$$

with: $\vec{R} = \vec{r} - \vec{r}\,'$

$$\vec{F}_{13} = \frac{(-3\mu C)(2\mu C)}{(4\pi)(8.854\times10^{-12}\frac{F}{m})5^3}[\hat{x}4-\hat{z}3]$$

$$= -\hat{x}\,1.73 + \hat{z}\,1.29\ mN$$

$$\vec{F}_{23} = \frac{(1\mu C)(2\mu C)}{(4\pi\epsilon)(20)^{3/2}}[\hat{x}4+\hat{z}2]$$

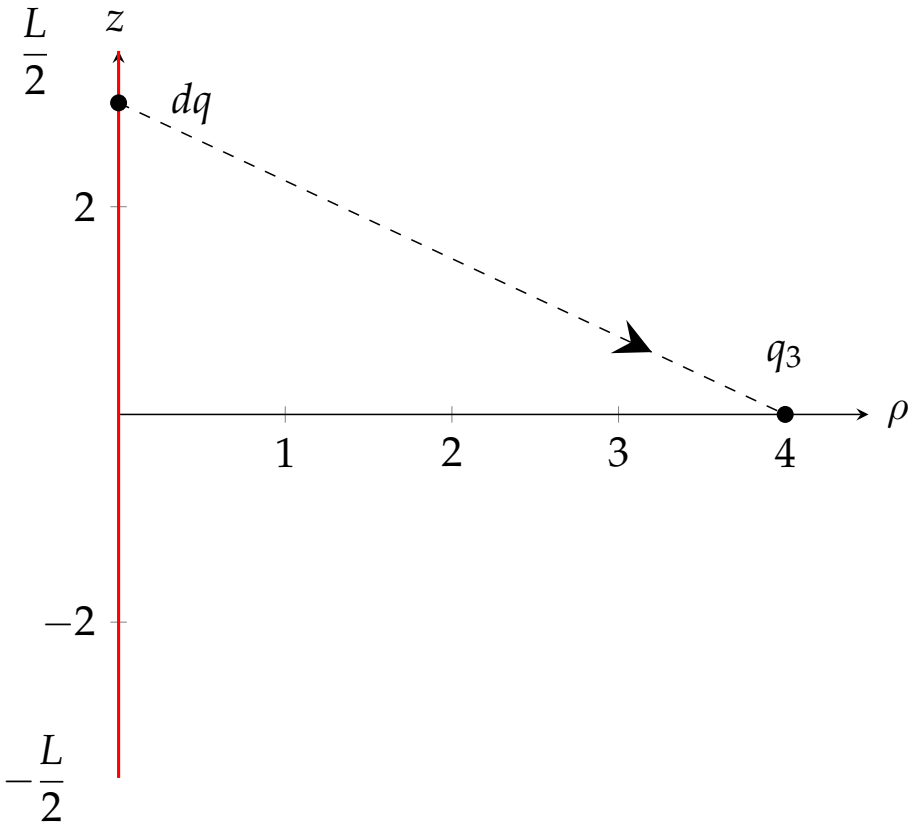
$$= -\hat{x}\,0.80 + \hat{z}\,0.4\ mN$$

$$\vec{F}_T = -\hat{x}\,0.93 + \hat{z}\,1.69\ mN$$

$$\vec{E}_3 = \frac{\vec{F}_T}{q_3}$$

# $\vec{E}$ by Direct Integration: Line Charge

$\int dE = E$ !



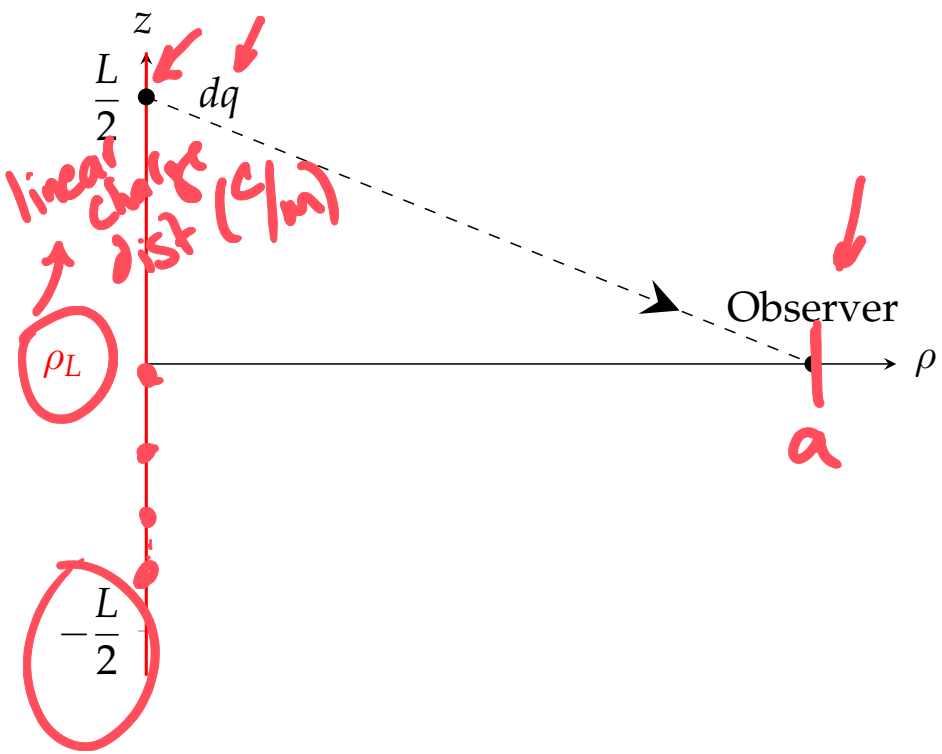

## Method

1. Draw picture
2. Write "Coulomb's Law":

$$d\vec{E} = \frac{dQ}{4\pi\varepsilon|R|^3}\vec{R}$$

3. Define $dQ$ in terms of charge distribution ($\rho$) and length ($d\ell$)
4. Define $\vec{R}$
5. Integrate to find $\vec{E}$

# $\vec{E}$ by Direct Integration: Line Charge




## Method

1. Draw picture
2. Write "Coulomb's Law":

$$d\vec{E} = \frac{dQ}{4\pi\varepsilon|R|^3}\vec{R}$$

3. Define $dQ =$
4. Define $\vec{R} =$

$$d\vec{E} =$$

# $\vec{E}$ BY DIRECT INTEGRATION: LINE CHARGE

$\hat{y}$ component

$$\hat{y}\int_{-L/2}^{L/2}\frac{\rho_L q}{4\pi\epsilon(a^2+z^2)^{3/2}}dz$$

$$\hat{y}\frac{\rho_L q}{4\pi\epsilon_0}\int_{-L/2}^{L/2}\frac{1}{(a^2+z^2)^{3/2}}dz$$

$$\hat{y}\frac{\rho_L q}{4\pi\epsilon_0}\left[\frac{z}{a^2\sqrt{z^2+a^2}}\right]_{z=-L/2}^{z=L/2}$$

$$\hat{y}\frac{\rho_L q}{4\pi\epsilon}\left[\frac{\frac{L}{2}}{a^2\sqrt{(\frac{L}{2})^2+a^2}}-\frac{-\frac{L}{2}}{a^2\sqrt{(\frac{L}{2})^2+a^2}}\right]$$

$$\boxed{\hat{y}\frac{\rho_L q L}{4\pi\epsilon a^2\sqrt{(\frac{L}{2})^2+a^2}}}$$

$\hat{z}$ component

$$-\hat{z}\int_{-L/2}^{L/2}\frac{\rho_L z}{4\pi\epsilon(a^2+z^2)^{3/2}}dz$$

$$=-\hat{z}\frac{\rho_L}{4\pi\epsilon}\int_{-L/2}^{L/2}\frac{z}{(a^2+z^2)^{3/2}}dz$$

Let $u=a^2+z^2$ $\quad du=2z\,dz$ $\quad \frac{1}{2}du=z\,dz$

$$\frac{1}{2}\int\frac{1}{u^{3/2}}du$$

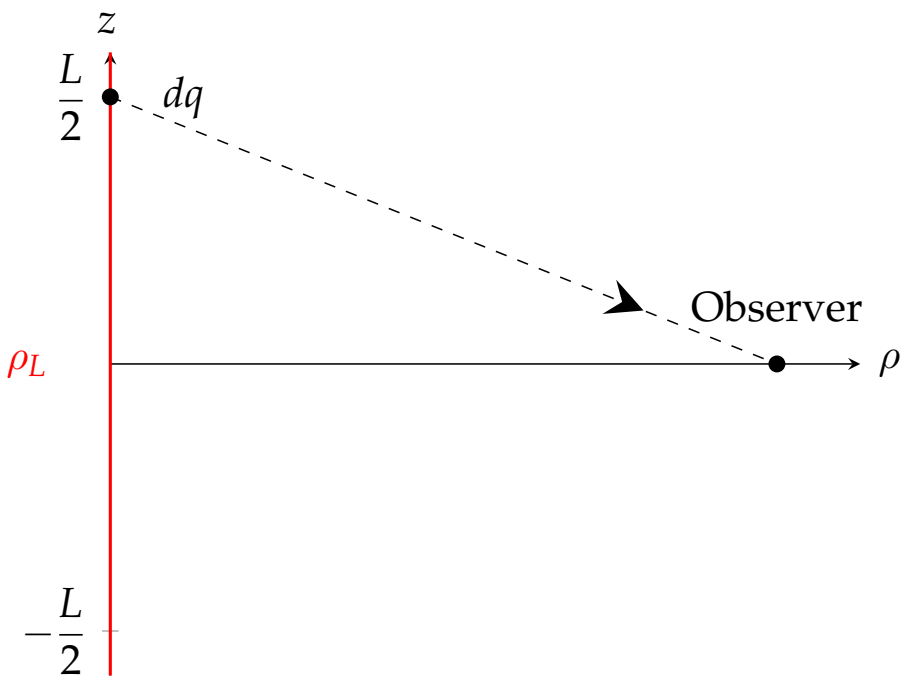
$$\frac{1}{2}(u^{-1/2})(-2)=-u^{-1/2}$$

$$=\left[-(a^2+z^2)^{-1/2}\right]_{-L/2}^{L/2}$$

$$=-\left[\frac{1}{(a^2+(\frac{L}{2})^2)^{1/2}}-\frac{1}{(a^2+(-\frac{L}{2})^2)^{1/2}}\right]$$

$$=0$$

# $\vec{E}$ by Direct Integration: Line Charge




## Result

$$\vec{E} = \hat{\boldsymbol{\rho}} \frac{\rho_L L}{4\pi\varepsilon\rho\sqrt{\rho^2 + \left(\frac{L}{2}\right)^2}}$$

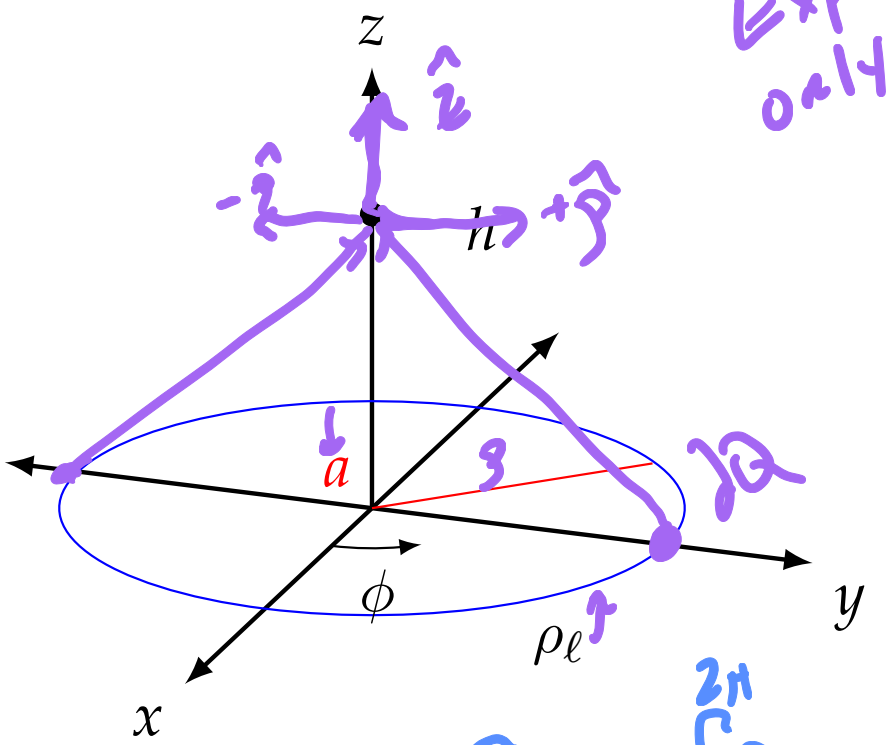
## Infinite Wire Approx

For $\frac{L}{2} >> \rho$

$$\vec{E} = \hat{\boldsymbol{\rho}} \frac{\rho_L L}{4\pi\varepsilon\rho\sqrt{\left(\frac{L}{2}\right)^2}} = \frac{\rho_L}{2\pi\varepsilon\rho}$$

# Wire Loop: Direct Integration

Find the Electric field at $z = h$ from a line charge carrying a charge distribution of $\rho_\ell$.



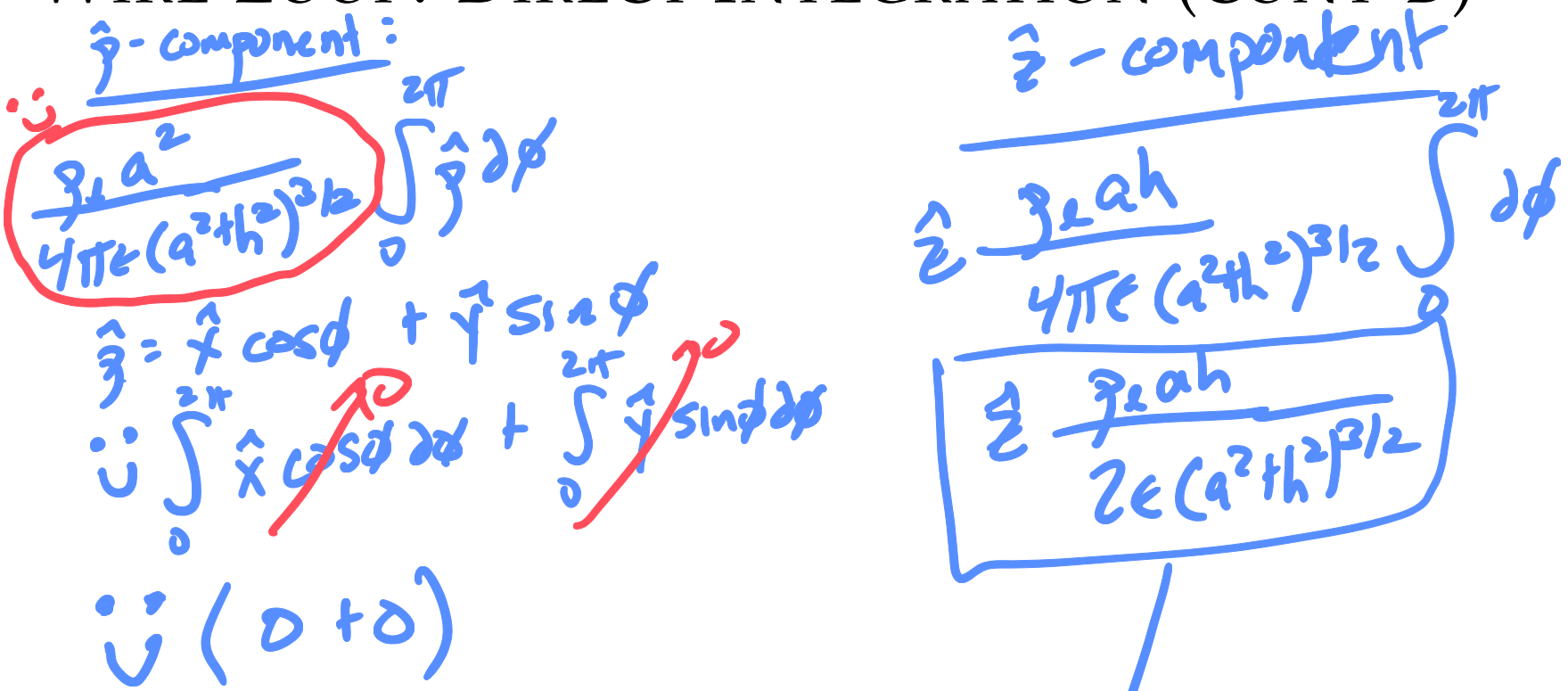

Expect only $\hat{z}$

## Method

1. Draw picture
2. Write "Coulomb's Law":
$$d\vec{E} = \frac{dQ}{4\pi\varepsilon|R|^3}\vec{R}$$
3. Define $dQ = \rho_\ell\, d\ell = \rho_\ell\, a\, d\phi$
4. Define $\vec{R} = \vec{r} = -\hat{\rho}a + \hat{z}h$

$$\int d\vec{E} = \int_0^{2\pi} \frac{\rho_\ell a}{4\pi\epsilon(a^2+h^2)^{3/2}} \left[-\hat{\rho}a + \hat{z}h\right] d\phi$$

# Wire Loop: Direct Integration (cont'd)

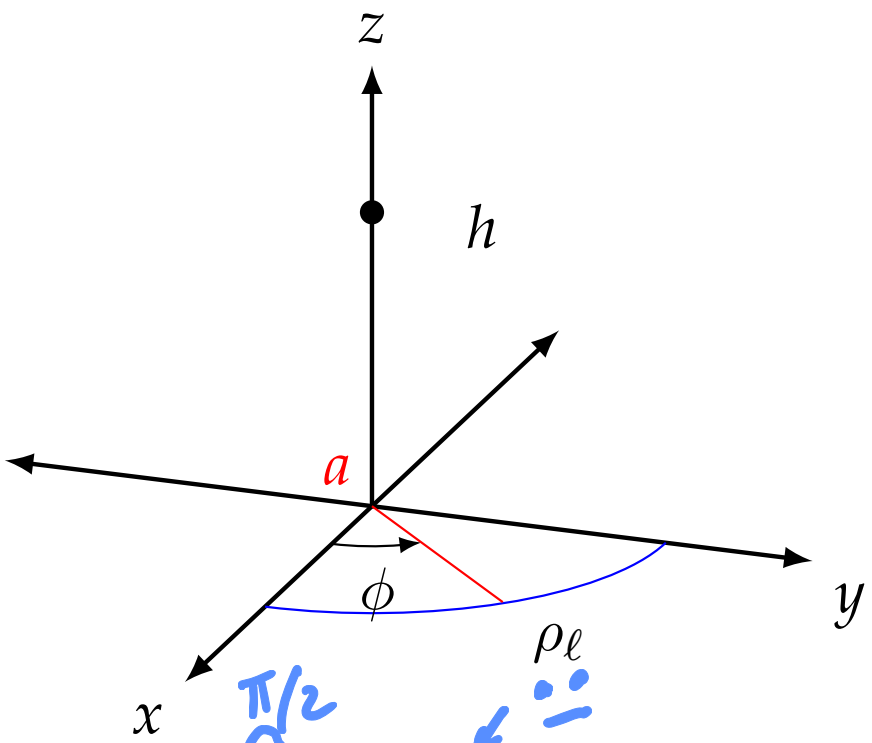



## Result

$$\vec{E} =$$

# Partial Line: Direct Integration

Find the Electric field at $z = h$ from a partial loop carrying a charge distribution of $\rho_\ell$.




## Method

1. Draw picture
2. Write "Coulomb's Law":
$$d\vec{E} = \frac{dQ}{4\pi\varepsilon|R|^3}\vec{R}$$
3. Define $dQ =$ $\rho_\ell dl = \rho_\ell a\, d\phi$
4. Define $\vec{R} =$
$d\vec{E} =$ _____

$$\vec{E} = \int_0^{\pi/2} \frac{\rho_\ell a}{4\pi\varepsilon(a^2+h^2)}\left[-\hat{x}a + \hat{z}h\right] d\phi$$

# Partial Line: Direct Integration

$\hat{x}$-component:

$$= \int_0^{\pi/2} -\hat{x}a\cos\phi\,d\phi - \int_0^{\pi/2} \hat{y}a\sin\phi\,d\phi$$

$$= -\left[\hat{x}a(\sin(\pi/2) - \sin(0)) + \hat{y}a\left(-\cos(\pi/2) + \cos(0)\right)\right]$$

$$= -\left[\hat{x}a + \hat{y}a\right]$$

$\hat{z}$-component

$$= h\int_0^{\pi/2} d\phi$$

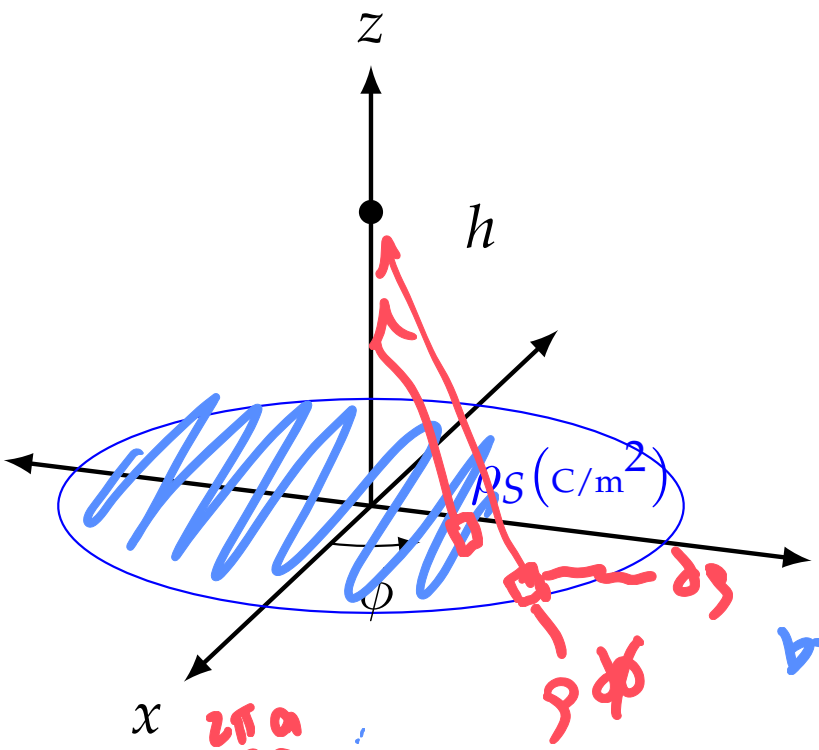
$$= \frac{\pi h}{2}\hat{z}$$

**Result**

$$\vec{E} = \frac{3\pi q}{4\pi\epsilon(a^2+h^2)^{3/2}}\left[-\hat{x}a - \hat{y}a + \hat{z}\frac{\pi h}{2}\right]$$

# Disk Charge: Direct Integration

Find the Electric field at $z = h$ from a disk carrying a charge distribution of $\rho_S$.




## Method

1. Draw picture
2. Write "Coulomb's Law":
$$d\vec{E} = \frac{dQ}{4\pi\varepsilon|R|^3}\vec{R}$$
3. Define $dQ =$ $\rho_s\,ds = \rho_s\,\rho\,d\phi\,d\rho$
4. Define $\vec{R} =$ $-\hat{\rho}\,\rho + \hat{z}\,h$
5. $d\vec{R} =$ _____

by symmetry

$$\iint d\vec{E} = \int_0^{2\pi}\int_0^a \frac{\rho_s\,\rho}{4\pi\varepsilon(\rho^2+h^2)^{3/2}}\left[-\hat{\rho}\,\rho + \hat{z}\,h\right]d\rho\,d\phi$$

# Disk Charge: Direct Integration (cont'd)

6. Integrate to find $\vec{E}$: $\hat{\rho} \to 0$ by symmetry

$\hat{z}$ - component

$$\frac{\rho_s h}{4\pi\epsilon} \int_0^{2\pi} d\phi \int_0^a \frac{\rho}{(\rho^2+h^2)^{3/2}} d\rho$$

$$\frac{\rho_s h}{2\epsilon} \int_0^a \frac{\rho}{(\rho^2+h^2)^{3/2}} d\rho$$

Let $u = \rho^2 + h^2$ $\quad \frac{du}{d\rho} = 2\rho$, $\quad \frac{1}{2}du = \rho d\rho$

$$\frac{1}{2}\int u^{-3/2} du = \frac{1}{2}(-2)\, u^{-1/2} = -\frac{1}{u^{1/2}} = -\frac{1}{(\rho^2+h^2)^{1/2}}\Big|_0^a$$

$$\to -\frac{\rho_s h}{2\epsilon(\rho^2+h^2)^{1/2}}\Big|_0^a$$

$$\frac{-\rho_s h}{2\epsilon}\left[\frac{1}{(a^2+h^2)^{1/2}} - \frac{1}{h}\right]$$

**Result**

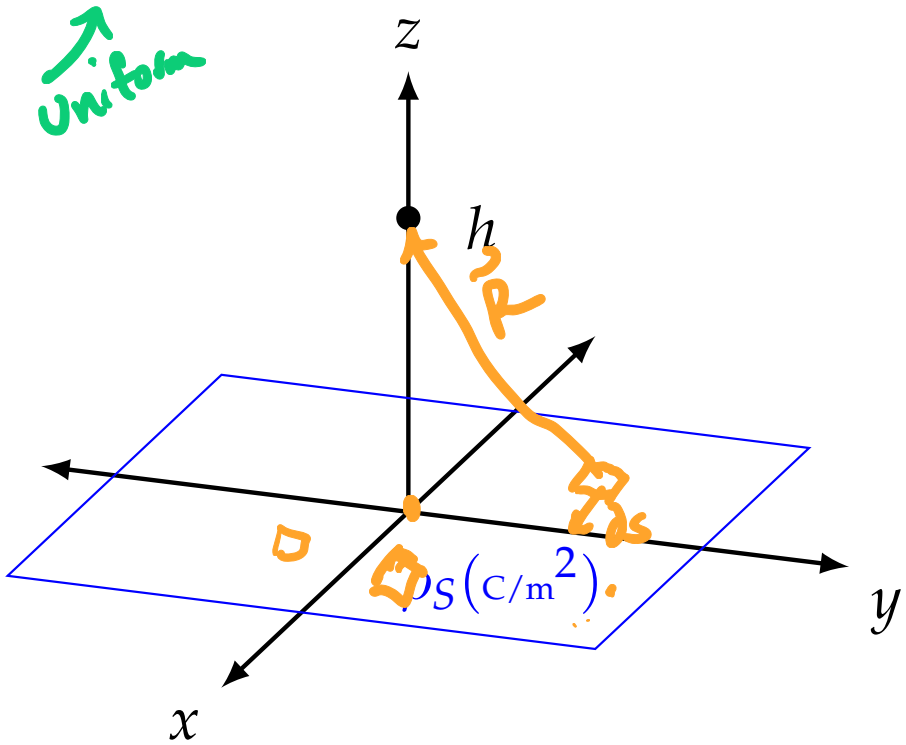
$$\frac{\rho_s h}{2\epsilon}\left[\frac{1}{h} - \frac{1}{(a^2+h^2)^{1/2}}\right]$$

**Infinite Disk ($h << a$)**

$$\frac{\rho_s}{2\epsilon}$$

# Sheet Surface Charge: Direct Integration

Find the Electric field at $z = h$ from a sheet carrying a charge distribution of $\rho_S$.




## Method

1. Draw picture
2. Write "Coulomb's Law":

$$d\vec{E} = \frac{dQ}{4\pi\varepsilon|R|^3}\vec{R}$$

3. Define $dQ = \rho_S\, dS = \rho_S\, dx\, dy$
4. Define $\vec{R} = +\hat{x}x - \hat{y}y + \hat{z}h$

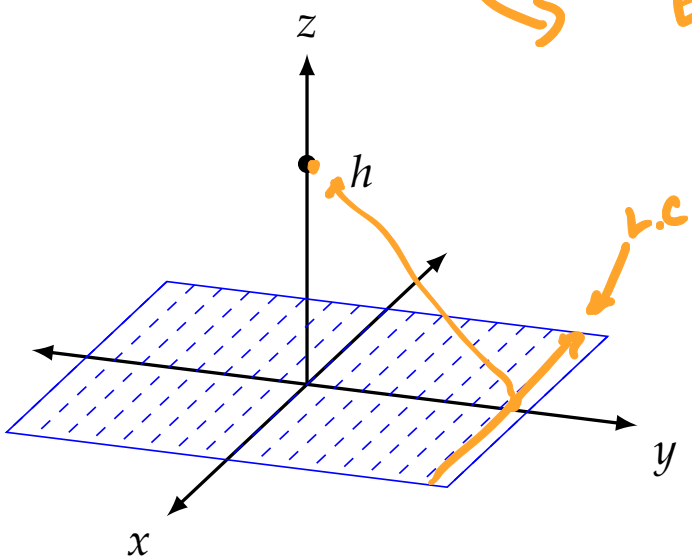
$$d\vec{E} = \frac{\rho_S}{4\pi\varepsilon(x^2+y^2+h^2)^{3/2}}\left[+\hat{x}x \pm \hat{y}y + \hat{z}h\right]dx\,dy$$

# Sheet Surface Charge: Direct Integration

Even when we notice the $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ components cancel (by symmetry),

$$\iint \frac{\rho_S}{(4\pi\varepsilon)\left(x^2+y^2+h^2\right)^{3/2}} \left(\pm\hat{\mathbf{x}}x \pm \hat{\mathbf{y}}y + \hat{\mathbf{z}}h\right) dxdy \longleftarrow \text{ gross...}$$

Another approach $\longrightarrow$ line charges (with $\rho_\ell$) oriented in $x$-direction...




$\vec{E}_{LC} = \hat{\rho}\frac{\rho_\ell}{2\pi\epsilon\rho}$ $\quad \hat{\rho}' = \frac{\pm\hat{y}y + \hat{z}h}{\sqrt{y^2+h^2}}$ $\quad \rho = \sqrt{y^2+h^2}$

L.C.

$$\vec{E}_{sc} = \int \frac{\rho_\ell}{2\pi\epsilon\sqrt{y^2+h^2}} \frac{\left[\pm\hat{y}y + \hat{z}h\right]}{\sqrt{y^2+h^2}} dy$$

$$= \lim_{A\to\infty} \frac{\rho_\ell}{2\pi\epsilon} \int_{-A}^{A} \frac{1}{y^2+h^2} \left[\pm\hat{y}y + \hat{z}h\right] dy$$

# Sheet Surface Charge: Direct Integration

$$\vec{E}_{sc} = \frac{\hat{z}\rho_s h}{2\pi\epsilon}\int_{-\infty}^{\infty}\frac{1}{y^2+h^2}\,dy = \hat{z}\frac{\rho_s}{2\pi}\left[\frac{1}{h}\tan\left(\frac{y}{h}\right)\right]_{-\infty}^{\infty}$$

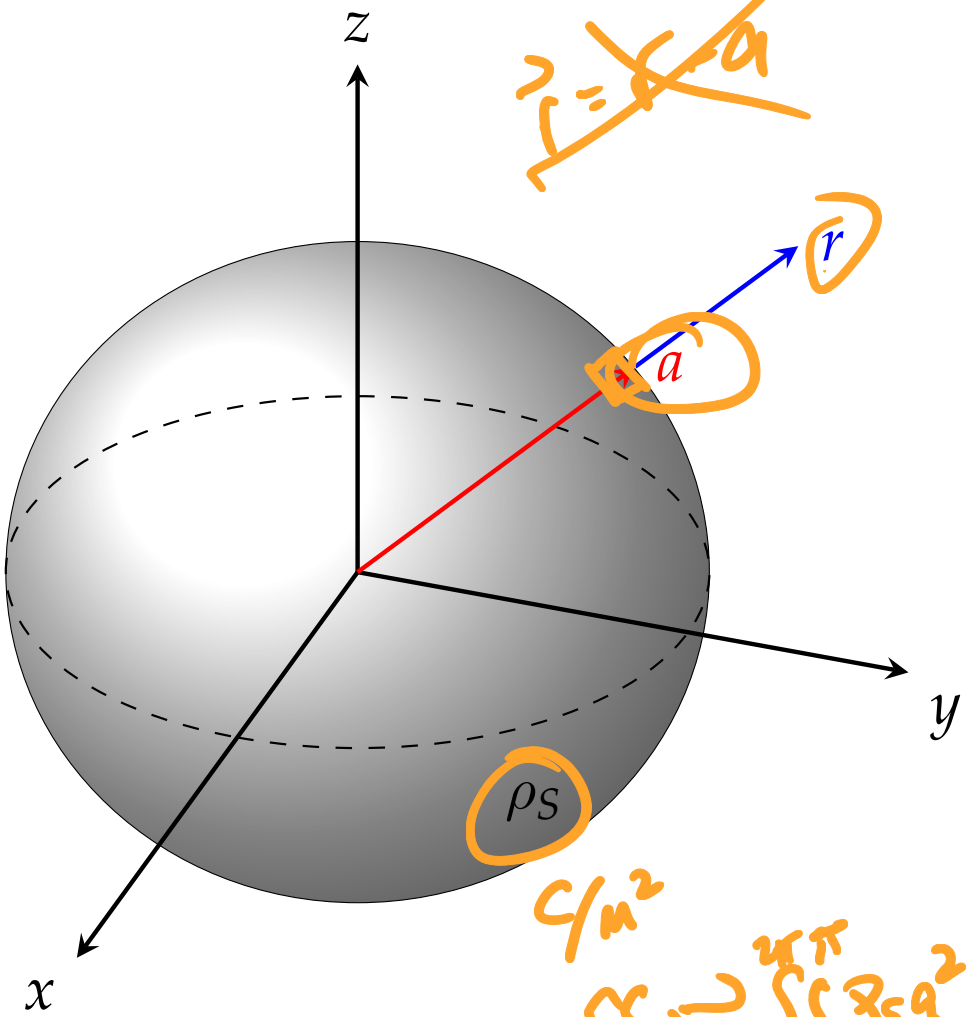
$$\hat{z}\frac{\rho_s h}{2\pi\epsilon}\left(\frac{\pi}{h}\right)$$

$$\vec{E}_{sc} = \hat{z}\frac{\rho_s}{2\epsilon}$$

**Result**

$$\vec{E} =$$

# Spherical Surface Charge




## Method

1. Draw picture
2. Write "Coulomb's Law":
$$d\vec{E} = \frac{dQ}{4\pi\varepsilon|R|^3}\vec{R}$$
3. Define $dQ =$ _____
4. Define $\vec{R} =$ _____

$$d\vec{E} = \text{_____}$$

# Spherical Surface Charge (cont'd)

$$\hat{r}\,\frac{\rho_s a^2}{4\pi\epsilon r^2}\int_0^{\pi}\sin\theta\,d\theta\int_0^{2\pi}d\phi$$

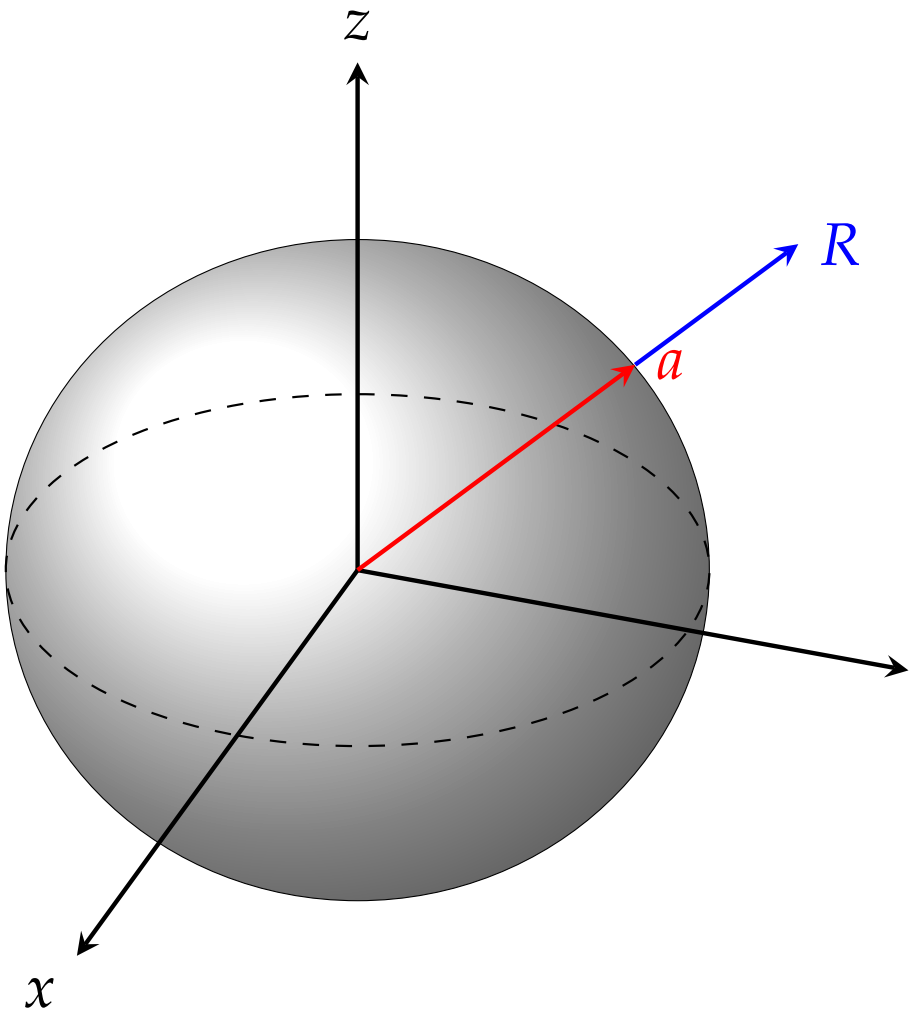
$$\int_0^{\pi}\sin\theta\,d\theta = 2 \qquad \int_0^{2\pi}d\phi = 2\pi$$

$$\vec{E} = \hat{r}\,\frac{\rho_s a^2}{\epsilon r^2}$$

## Result

$$\vec{E} = \underline{\qquad\qquad\qquad\qquad\qquad\qquad}$$

# Your Turn: Spherical Volume Charge




## Method

1. Draw picture
2. Write "Coulomb's Law":

$$d\vec{E} = \frac{dQ}{4\pi\varepsilon|R|^3}\vec{R}$$

3. Define $dQ =$
4. Define $\vec{R} =$

$$d\vec{E} = _____$$

# Spherical Volume Charge (cont'd)

**Result**

$$\vec{E} = \hat{\mathbf{r}} \frac{\rho_V a^3}{3\varepsilon R^3}$$